

PSG COLLEGE OF TECHNOLOGY



Industrial Training Report On 230/110 kV Substation



SUBMITTED BY

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TRAINING DURATION	: ONE WEEK
DATE	: 22 JUNE 26 - 26 JUNE 26

CERTIFICATE

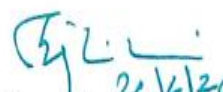
[Official TANTRANSCO Training Certificate]

TAMILNADU TRANSMISSION CORPORATION LIMITED

Office of the
Executive Engineer / Operation
230 / 110 KV Sub-Station,
Samayanallur

INPLANT TRAINING CERTIFICATE **TO WHOMSOEVER IT MAY CONCERN**

This is to certify that, Thiru. **N.Dinesh Kailash**, student of PSG College of Technology, Coimbatore., (B.E-EEE) had undergone "**IN-PLANT TRAINING**" at Samayanallur 230 / 110 KV Sub-Station (TANTRANSCO) for the period from 22.06.2026 to 26.06.2026. He had completed the training satisfactorily.


Executive Engineer / Operation
230/110 KV Sub-Station,
Samayanallur

ACKNOWLEDGEMENT

I express my sincere gratitude to Tamil Nadu Transmission Corporation Limited (TANTRANSCO), 230/110 kV Substation, Samayanallur, for providing me with the opportunity to undergo a one-week industrial training programme from 22 June 2026 to 26 June 2026. The training provided me with valuable practical exposure to the operation, maintenance, and protection of high-voltage substations.

I would like to extend my heartfelt thanks to the Executive Engineer (Operation) and all the engineers and technical staff of the substation for their continuous guidance, encouragement, and for patiently explaining the various equipment, protection systems, and operational procedures during the training.

I am also grateful to the Department of Electrical and Electronics Engineering, PSG College of Technology, for granting me the opportunity to undergo this industrial training as a part of my academic curriculum.

Finally, I express my sincere thanks to everyone who directly and indirectly contributed to the successful completion of my industrial training and the preparation of this report.

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INTRODUCTION

Industrial training is an essential part of engineering education as it bridges the gap between theoretical knowledge and practical applications. It provides students with an opportunity to observe industrial practices, understand the operation of electrical equipment, and gain exposure to real-world engineering environments. Through industrial training, students develop technical knowledge, problem-solving skills, and an understanding of safety procedures followed in industries.

As a part of the Bachelor of Engineering (B.E.) curriculum in Electrical and Electronics Engineering at **PSG College of Technology**, I underwent a **one-week industrial training** at the **230/110 kV Substation, Samayanallur, operated by the Tamil Nadu Transmission Corporation Limited (TANTRANSCO)**, from **22 June 2026 to 26 June 2026**.

The training provided valuable practical exposure to the functioning of an Extra High Voltage (EHV) substation. During the training, I studied the construction, operation, and maintenance of major substation equipment such as **power transformers, circuit breakers, isolators, current transformers (CTs), potential transformers (PTs), lightning arresters, busbars, and protection relays**. I also gained an understanding of switching operations, safety practices, preventive maintenance, and protection schemes adopted in high-voltage substations.

This report presents the knowledge and practical experience gained during the training. It includes an overview of TANTRANSCO, the 230/110 kV substation, major electrical equipment, protection systems, operation and maintenance practices, safety measures, and the key learning outcomes achieved during the industrial training.

ABOUT TANTRANSCO

The **Tamil Nadu Transmission Corporation Limited (TANTRANSCO)** is a wholly owned subsidiary of **Tamil Nadu Generation and Distribution Corporation Limited (TANGEDCO)**. It was established on **1 November 2010** after the restructuring of the former Tamil Nadu Electricity Board (TNEB). TANTRANSCO is responsible for the transmission of electrical power throughout the state of Tamil Nadu and plays a vital role in ensuring a reliable and efficient electricity supply.

The primary responsibility of TANTRANSCO is to transmit electrical energy generated from various sources, including thermal, hydro, nuclear, solar, and wind power stations, to distribution substations across the state. It operates and maintains Extra High Voltage (EHV) transmission lines and substations, ensuring that electricity is delivered safely and efficiently to consumers through the distribution network.

TANTRANSCO manages an extensive network of transmission infrastructure consisting of **765 kV, 400 kV, 230 kV, 110 kV, and 66 kV** transmission lines and substations. The organization continuously upgrades its transmission facilities to improve system reliability, reduce transmission losses, and meet the increasing demand for electricity in Tamil Nadu.

The corporation is also responsible for planning, constructing, operating, and maintaining transmission systems while ensuring compliance with national safety standards and regulatory requirements. Modern technologies such as **Supervisory Control and Data Acquisition (SCADA)** systems, digital protection relays, remote monitoring, and automation are employed to improve operational efficiency and enhance the reliability of the power system.

During my industrial training, I visited the **230/110 kV Substation, Samayanallur**, operated by TANTRANSCO. The training provided practical exposure to the construction, operation, maintenance, and protection of major substation equipment such as power transformers, circuit breakers, isolators, current transformers (CTs), potential transformers (PTs), busbars, lightning arresters, and protective relays. The experience helped me understand the practical implementation of concepts studied in power systems and electrical engineering courses.

ABOUT THE 230/110 kV SUBSTATION

The **230/110 kV Substation at Samayanallur**, operated by **Tamil Nadu Transmission Corporation Limited (TANTRANSCO)**, is an Extra High Voltage (EHV) transmission substation responsible for receiving electrical power at **230 kV**, stepping it down to **110 kV**, and transmitting it to various load centers through outgoing transmission lines. The substation plays a vital role in maintaining the stability, reliability, and continuity of the power transmission network.

The substation receives electrical power through three **230 kV incoming transmission lines**, namely **Alagar Kovil Line**, **Pasumalai Line**, and **Kinnimangalam Line**. Each incoming line is connected through essential switchyard equipment such as **Current Transformers (CTs)**, **Potential Transformers (PTs)**, **Circuit Breakers (CBs)**, and **Isolators** before being connected to the **230 kV double busbar arrangement**. This arrangement provides operational flexibility and allows maintenance work to be carried out without interrupting the entire power supply.

The received electrical power is stepped down from **230 kV to 110 kV** using **two 100 MVA power transformers**. After voltage transformation, the power is transferred to the **110 kV busbar** and supplied through outgoing feeders to nearby substations, including **Pasumalai, Arasadi, Sembatty, Manikampatty, Vadipatti, and Alagar Kovil**.

The substation is also provided with **two 11 kV station service transformers** to meet its internal auxiliary power requirements. These transformers supply power for lighting, battery chargers, air-conditioning, relay panels, control equipment, and other auxiliary loads required for the operation of the substation.

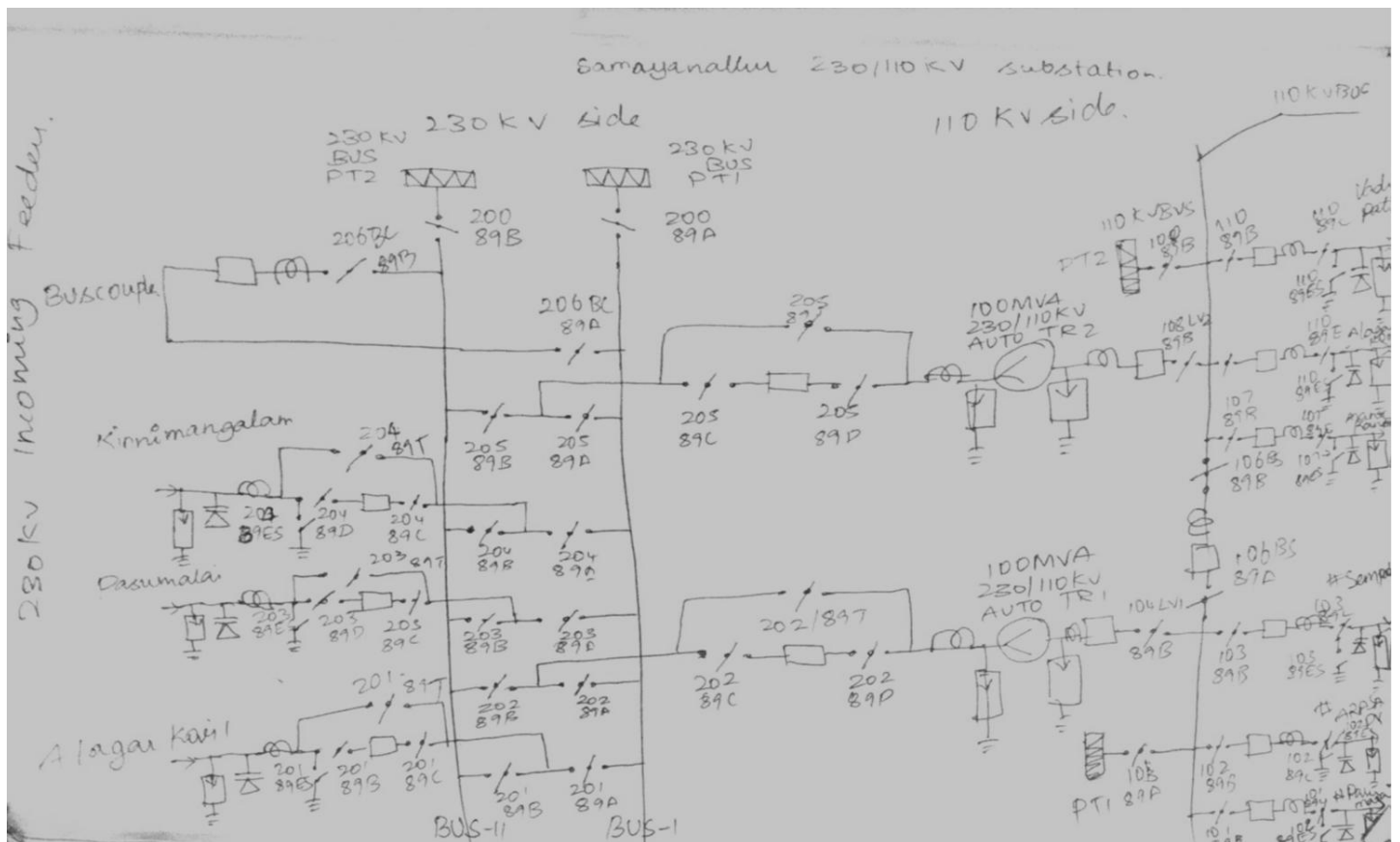
To ensure uninterrupted operation during AC supply failures, the substation is equipped with **220 V DC and 48 V DC battery systems**. These battery banks provide reliable DC power for protection relays, circuit breaker operation, communication systems, and control circuits.

The relay and control room houses protection relays, control panels, synchronizing panels, and monitoring equipment. The room is maintained under controlled temperature conditions using air-conditioning to ensure reliable operation of sensitive electronic devices. The substation is also equipped with a **Supervisory Control and Data Acquisition (SCADA)** system, which enables continuous monitoring and remote operation of various substation equipment from the control room.

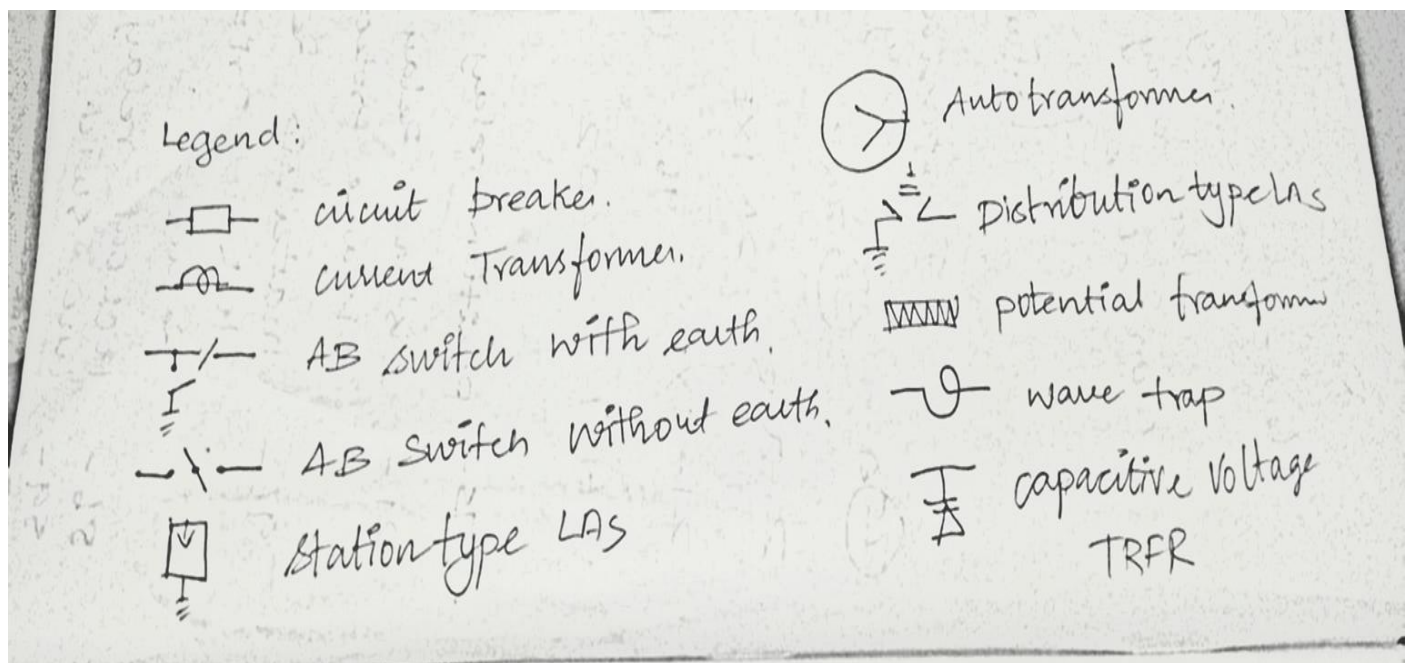
During the industrial training, I observed the arrangement and operation of the switchyard, power transformers, auxiliary power systems, relay panels, battery banks, and SCADA system.

The visit provided valuable practical exposure to the working principles and operation of an Extra High Voltage transmission substation.

this is diagram is drawn from given single line diagram



single line diagram of 230/110kV substation



SWITCHYARD OBSERVATION

During the industrial training at the **230/110 kV Samayanallur Substation**, I observed the switchyard where electrical power is received, controlled, and distributed.

The substation receives **230 kV power** through three incoming transmission lines:

- Alagar Kovil
- Pasumalai
- Kinnimangalam

Each incoming feeder passes through **Current Transformer (CT), Potential Transformer (PT), Circuit Breaker (CB), and Isolator (DS)** before being connected to the **230 kV double busbar**. A bus coupler is provided between the two busbars to improve operational flexibility and reliability.

The power is then supplied to **two 100 MVA, 230/110 kV power transformers**, where the voltage is stepped down to **110 kV**.

From the **110 kV bus**, power is transmitted through the following outgoing feeders:

- Pasumalai
- Arasadi
- Sembatty
- Manikampatty
- Vadipatti
- Alagar Kovil

During the visit, I observed the arrangement of the switchyard equipment and understood the flow of electrical power from the incoming feeders to the outgoing feeders through the transformers. The engineers explained the purpose of the CTs, PTs, circuit breakers, isolators, and busbars used in the switchyard.



POWER TRANSFORMER OBSERVATION

During the industrial training at the **230/110 kV Samayanallur Substation**, I observed the **230/110 kV power transformers** from a safe distance because the substation was in operation. The engineers explained the construction, operation, and cooling system of the transformer.

The substation is equipped with **two 100 MVA, 230/110 kV oil-immersed power transformers**. Their primary function is to step down the transmission voltage from **230 kV to 110 kV** for further distribution.

The engineers explained that the transformer is filled with **transformer oil**, which performs two important functions:

- It acts as an **insulating medium** between internal components.
- It helps **remove the heat** generated in the transformer windings and core during operation.

The heated oil rises to the top of the transformer and flows through the **radiators**, where it loses heat to the surrounding air. After cooling, the oil returns to the transformer tank, creating a continuous cooling cycle that helps maintain a safe operating temperature.

The transformer is also provided with a **conservator tank**, which accommodates the expansion and contraction of transformer oil due to temperature changes.

The engineers showed the **silica gel breather (moisture absorber)** connected to the conservator tank. The silica gel absorbs moisture from the air entering the conservator, preventing water from mixing with the transformer oil. They explained that the silica gel is normally **blue** (or orange, depending on the type used) and changes colour when it absorbs moisture. During the visit, we were informed that **if the silica gel changes to pink, it should be replaced or regenerated** to maintain the quality of the transformer oil.

Although I could not inspect the transformer closely due to safety restrictions, the explanations provided by the engineers helped me understand the working principle, cooling method, and maintenance requirements of an oil-immersed power transformer.



STATION AUXILIARY SUPPLY AND BATTERY SYSTEM

The substation requires a continuous power supply for its control equipment, protection relays, lighting, communication systems, and office operations. During the industrial training, I observed the station auxiliary supply and battery system used to provide uninterrupted power to these essential services.

The substation is equipped with **two 11 kV station service transformers**. These transformers supply power to the station auxiliaries. The AC supply is fed to **battery chargers (rectifiers)**, which convert it into **220 V DC** and **48 V DC** for various applications within the substation.

The **220 V DC battery bank** consists of **110 batteries**, each rated at **2 V** with a capacity of **300 Ah**. This battery bank is mainly used for operating protection relays, circuit breaker tripping and closing circuits, control panels, and other essential control equipment. The DC supply ensures that these systems continue to function even if the incoming AC supply fails.

The substation also has a **48 V DC battery bank** consisting of **24 batteries**, each rated at **2 V** with a capacity of **400 Ah**. This battery bank supplies power to communication equipment and other low-voltage DC loads used within the control room and office.

The engineers explained that the batteries are continuously charged during normal operation. In the event of a power failure, the battery banks automatically supply DC power, allowing the protection and control systems to operate without interruption. This uninterrupted DC supply is essential for the safe and reliable operation of the substation.

During the visit, I observed the battery room, battery banks, battery chargers, and the station auxiliary supply arrangement. The engineers explained the importance of maintaining the batteries properly to ensure reliable operation during emergency conditions.

RELAY ROOM AND SCADA

During the industrial training, I visited the **relay room** and **SCADA control room**, where the engineers explained the protection and control system used in the substation.

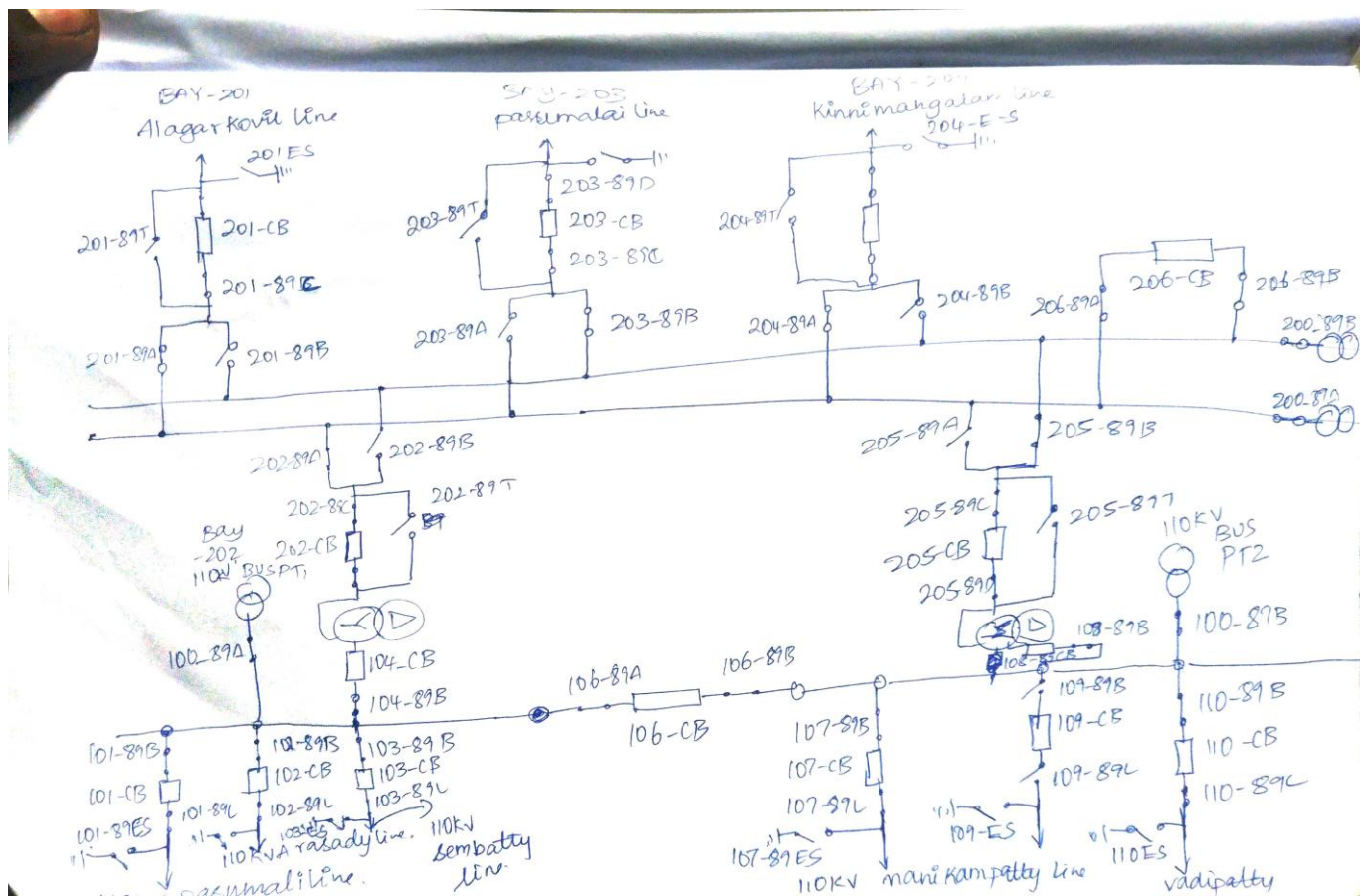
The relay room contains **numerical relays**, which are microprocessor-based programmable relays used to protect the transmission lines and transformers. The engineers explained that these relays receive current and voltage signals from the **Current Transformer (CT)** and **Capacitive Voltage Transformer (CVT)**. Based on the programmed settings, the relay detects abnormal conditions and sends a trip signal to the circuit breaker to isolate the faulty section.

During the visit, I observed different protection relays and learned about several ANSI relay numbers such as **21 (Distance Protection)**, **25 (Synchronism Check)**, **27 (Under Voltage)**, **50 (Instantaneous Overcurrent)**, **51 (Time Overcurrent)**, **59 (Over Voltage)**, **86 (Lockout Relay)**, **87 (Differential Protection)**, and **87T (Transformer Differential Protection)**. The engineers also explained relay indications such as **DEF, SOTF, VT Fail, BBC Alarm, Carrier Receive, DT Receive, CH-1 Fail, Ground Fault, Zone 1, Zone 2, Zone 3, Zone 4, and R, Y, B phase line faults**.

The engineers demonstrated the **SCADA (Supervisory Control and Data Acquisition)** system used to monitor and control the substation. They explained that in earlier days, circuit breakers were operated manually in the switchyard during maintenance or fault conditions. Later, control rooms were introduced so that operators could control circuit breakers from inside the station. With the introduction of SCADA, substation operation has become more convenient, reliable, and efficient.

Using the SCADA system, operators can monitor the status of circuit breakers and isolators, observe voltage and current values received from the CVTs and CTs, monitor power received and power delivered, view alarms and fault indications, and operate circuit breakers remotely from the control room. The engineers also explained that **PLC and SCADA** work together to improve automation and reduce manual operation in the substation.

During the visit, I gained a better understanding of modern protection systems, numerical relays, and SCADA-based monitoring and control, which play an important role in the safe and reliable operation of the substation.



This is our diagram from Scada system which shown in that substation.

KEY LEARNING AND CONCLUSION

The one-week industrial training at the **230/110 kV Samayanallur Substation** provided me with valuable practical knowledge about the operation of a transmission substation. It helped me understand how electrical power is received, transformed, protected, and transmitted safely.

During the training, I learned about the layout of the switchyard, the function of current transformers (CTs), potential transformers (PTs), circuit breakers, isolators, lightning arresters, and busbars. I also understood the importance of oil-immersed power transformers, transformer oil cooling, the conservator tank, and the silica gel breather.

The engineers explained the station auxiliary supply, battery charging system, and the importance of the **220 V DC** and **48 V DC** battery banks in supplying uninterrupted power to protection and control equipment during power failures.

I also gained knowledge about numerical relays, protection schemes, and the operation of the SCADA system used for monitoring and controlling the substation remotely. This training

helped me connect the theoretical concepts learned in college with their practical implementation in a real transmission substation.

Overall, the industrial training improved my technical knowledge, practical understanding, and interest in power system engineering. It was a valuable learning experience that will be beneficial for my future studies and career.

REFERENCES

1. TANTRANSCO – Operation and Maintenance Manual (Reference from Industrial Training).
2. Information and explanations provided by the engineers and staff of the **230/110 kV Samayanallur Substation** during the industrial training.